Chapter 1. Introduction and Elementary Concepts

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1 Collection of Survey Data

1.1 Primary and secondary data

Definition 1.1

- **Primary data**: data collected by the investigator directly from the source.
- **Secondary data**: data that has already been collected by some agency or individual and is available in published or unpublished records.

Primary data are collected for the purpose at hand, so they fit the target of the study; secondary data are cheaper but were collected for some other purpose and may not match the target exactly. Primary data can be collected by

- direct personal interviews, or interviews by enumerators,
- questionnaires sent by mail,
- telephone interviews.

1.2 Framing a questionnaire

The person framing the questionnaire should know the field of inquiry in detail. Points to keep in mind:

- The surveyor introduces themselves and states the objective of the survey.
- Questions are clear, unambiguous and to the point; leading questions are avoided.
- Questions affecting the prestige or sentiments of respondents, and questions requiring calculations, are avoided.
- Precise instructions for filling in the questionnaire, including units of measurement, are given.
- The questionnaire is concise. Every question costs response quality, so the number of questions should be kept close to the number of quantities (parameters) one actually wants to estimate.
- The layout is visually appealing.

1.3 Technical terms

Definition 1.2

- An **element** is a unit for which information is sought.
- The **population** (universe) is the aggregate of elements about which inference is to be made.
- **Sampling units** are non-overlapping collections of population elements.
- The **sampling frame** is a list of all sampling units in the population.
- A **sample** is a subset of the population selected from the frame to draw inferences about population characteristics.

Collecting information on every unit of the population is a **complete enumeration** or **census**. Often a census is impossible (the target population is not fully accessible in practice), and a sample survey has advantages over it: greater speed, more detailed information, reduced cost and, perhaps surprisingly, greater accuracy, since a small survey can be run with trained staff and careful follow-up, so non-sampling errors are smaller.

1.4 Probability and non-probability sampling

The method used to select the sample is the **sampling procedure**.

- **Probability sampling**: units are selected by a probability mechanism. Only these procedures are considered in this course.
- **Non-probability sampling**: units are selected without a probability mechanism (convenience, quota, judgement); such procedures are subject to human bias.

1.5 With and without replacement

Definition 1.3

- **With replacement (WR)**: units are drawn one by one, and the unit selected at each draw is returned to the population before the next draw. If order matters, there are N^n possible samples.
- **Without replacement (WOR)**: the selected unit is not returned before the next draw. There are $\binom{N}{n}$ possible (unordered) samples.

Example 1.4: Birth weights

Four children A, B, C, D have birth weights $5.5, 8.0, 6.5, 7.0$ pounds. The WR samples of size 2 (ordered) are the $4^2 = 16$ pairs

$$(A,A),(A,B),(A,C),(A,D),(B,A),\ldots,(D,D),$$

whereas WOR gives the $\binom{4}{2} = 6$ samples $\{A,B\},\{A,C\},\{A,D\},\{B,C\},\{B,D\},\{C,D\}$.

In R: `sample(n)` is a random permutation of $1,\ldots,n$; `sample(x)` permutes x; `sample(x, n)` draws n items WOR; `sample(x, n, replace=T)` draws WR; `sample(x, replace=T)` is a bootstrap sample.

1.6 Planning and execution of a sample survey

Objectives → population to be studied → sampling unit → sampling frame → sample selection (sample size and procedure) → method of collecting information → pilot survey → handling of nonresponse → organization of fieldwork → analysis of data and report.

1.7 Dense and sparse models

Remark 1.5: Modeling view of sampling designs

Under SRS the data are modeled as $y_i = \mu + \varepsilon_i$ $(i = 1,\ldots,n)$ and $\bar{y} = \frac{1}{n}\sum y_i$ uses every observation for the single parameter μ: a *dense* model. If instead

$$y_i = \mu + \varepsilon_i \ (i = 1,\ldots,n_1), \qquad y_j = \varepsilon_j \ (j = 1,\ldots,n - n_1),$$

only part of the data carries information on μ (a *sparse* model), and estimation must treat the two groups differently. Stratified sampling (Ch. 3) uses $y_i = \mu_1 + \varepsilon_i$ and $y_j = \mu_2 + \varepsilon_j$: every observation is informative, but about its own stratum mean.

2 Elementary Concepts

2.1 Expectation, variance, covariance

If x takes values $x_1,\ldots,x_k$ with probabilities $p_1,\ldots,p_k$,

$$\mathrm{E}(x) = \sum_{i=1}^{k} p_i x_i, \qquad \mathrm{Var}(x) = \mathrm{E}[x - \mathrm{E}(x)]^2 = \mathrm{E}(x^2) - [\mathrm{E}(x)]^2,$$

$$\mathrm{Cov}(x,y) = \mathrm{E}[(x - \mathrm{E}x)(y - \mathrm{E}y)] = \mathrm{E}(xy) - \mathrm{E}(x)\mathrm{E}(y),$$

$$\mathrm{Var}\Big(\sum_{i=1}^{k} a_i x_i\Big) = \sum_{i=1}^{k} a_i^2 \mathrm{Var}(x_i) + 2\sum_{i<j} a_i a_j \mathrm{Cov}(x_i, x_j).$$

2.2 Parameters and statistics

Definition 1.6

A **parameter** is a real-valued function of the variable values of all population units. A **statistic** is a real-valued function of the sample values; a statistic used to estimate a parameter is an **estimator**.

For population values $Y_1, \dots, Y_N$ and sample values $y_1, \dots, y_n$:

$$\mu = \frac{1}{N}\sum_{i=1}^{N} Y_i, \quad \sigma^2 = \frac{1}{N}\sum_{i=1}^{N}(Y_i-\mu)^2, \quad S^2 = \frac{1}{N-1}\sum_{i=1}^{N}(Y_i-\mu)^2 = \frac{N}{N-1}\sigma^2,$$

$$\bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i, \qquad s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(y_i-\bar{y})^2.$$

The finite-population variance S^2 (divisor $N-1$) is the natural target of s^2 under SRS without replacement (Ch. 2), and $(N-1)S^2 = N\sigma^2$.

2.3 Bias, variance and MSE

Definition 1.7

- $\hat{\theta}$ is **unbiased** for θ if $\mathrm{E}(\hat{\theta}) = \theta$. Otherwise its **bias** is $B(\hat{\theta}) = \mathrm{E}(\hat{\theta}) - \theta$ and its **relative bias** is $RB(\hat{\theta}) = B(\hat{\theta})/\theta$.
- Sampling variance $\mathrm{Var}(\hat{\theta}) = \mathrm{E}[\hat{\theta} - \mathrm{E}\hat{\theta}]^2$; standard error $SE(\hat{\theta}) = \sqrt{\mathrm{Var}(\hat{\theta})}$.
- Mean squared error $\mathrm{MSE}(\hat{\theta}) = \mathrm{E}(\hat{\theta} - \theta)^2$.
- The **relative efficiency** of $\hat{\theta}_2$ with respect to $\hat{\theta}_1$ is $\mathrm{RE}(\hat{\theta}_1, \hat{\theta}_2) = \mathrm{MSE}(\hat{\theta}_1)/\mathrm{MSE}(\hat{\theta}_2)$; if it exceeds 1, $\hat{\theta}_2$ is more efficient.

Theorem 1.8: Bias–variance decomposition

$\mathrm{MSE}(\hat{\theta}) = \mathrm{Var}(\hat{\theta}) + [B(\hat{\theta})]^2$.

Proof.

$$\begin{aligned}\mathrm{E}(\hat{\theta}-\theta)^2 &= \mathrm{E}\big[(\hat{\theta}-\mathrm{E}\hat{\theta}) + (\mathrm{E}\hat{\theta}-\theta)\big]^2 \\ &= \mathrm{E}(\hat{\theta}-\mathrm{E}\hat{\theta})^2 + (\mathrm{E}\hat{\theta}-\theta)^2 + 2(\mathrm{E}\hat{\theta}-\theta)\,\mathrm{E}(\hat{\theta}-\mathrm{E}\hat{\theta}),\end{aligned}$$

and the last expectation is zero. ∎

2.4 Confidence intervals and the bound on the error

For a desired probability $1-\alpha$ we look for a bound B with

$$P(|\hat{\theta}-\theta| \le B) = 1-\alpha \iff P(\hat{\theta}-B \le \theta \le \hat{\theta}+B) = 1-\alpha,$$

so $(\hat{\theta}-B, \hat{\theta}+B)$ is a confidence interval with confidence coefficient $1-\alpha$.

- **Normal case**: if $\hat{\theta}$ is (approximately) normal and unbiased, $B = z_{\alpha/2}\sigma_{\hat{\theta}}$. For $1-\alpha = 0.95$, $z_{0.025} = 1.96 \approx 2$, so $B = 2\sigma_{\hat{\theta}}$, the "2-SD bound on the error of estimation" used throughout the course.
- **Any distribution**: by Chebyshev's inequality $P(|\hat{\theta}-\theta| \le 2\sigma_{\hat{\theta}}) \ge 1 - \frac{1}{4} = 0.75$.

2.5 Sample size determination

Only the investigator can say how much precision is needed. It is expressed as $P(|\hat{\theta}-\theta| \le B) = 1-\alpha$, where B is the **tolerable error** (margin of error). For surveys of people measuring a proportion, $B = 0.03$ and $\alpha = 0.05$ are typical.

Example 1.9: Opinion polls

With the normal approximation and $p = 1/2$ (the worst case),

$$B = 1.96\sqrt{\frac{p(1-p)}{n}} \le 1.96\sqrt{\frac{1}{4n}}.$$

Setting $B = 0.03$ gives $n = (1.96)^2/(4{\cdot}0.03^2) = 1067.1$, so about 1,068 respondents; this is why national polls typically report a margin of error of $\pm 3\%$ with about a thousand respondents.

3 Selected exercises

Motivation. These items practise the basic tools of the chapter: the probability guarantee behind the 2-SD bound, an exact sampling distribution computed by enumeration, and the fact that a biased estimator can beat an unbiased one in MSE.

Exercise 1.1 (Chebyshev guarantee for the bound on the error)

Let $\hat{\theta}$ be an estimator of θ with finite variance, and let $k > 0$.
(a) If $\hat{\theta}$ is unbiased with standard error $\sigma_{\hat{\theta}} > 0$, show that $P(|\hat{\theta} - \theta| \le k\sigma_{\hat{\theta}}) \ge 1 - 1/k^2$.
(b) Without assuming unbiasedness, show that $P(|\hat{\theta}-\theta| \le k\sqrt{\mathrm{MSE}(\hat{\theta})}) \ge 1-1/k^2$ whenever $\mathrm{MSE}(\hat{\theta}) > 0$, and explain why the 2-SD bound of a biased estimator should be built from $\sqrt{\mathrm{MSE}}$ rather than from the standard error.

Solution. [Supplement] Both parts follow from one inequality.

Step 1. (Markov's inequality.) For a random variable $U \ge 0$ and $a > 0$, $a\,\mathbf{1}\{U \ge a\} \le U$ pointwise; taking expectations gives $P(U \ge a) \le \mathrm{E}(U)/a$.

Step 2. Apply Step 1 to $U = (\hat{\theta} - \theta)^2$ and $a = k^2\mathrm{E}(\hat{\theta} - \theta)^2 = k^2\mathrm{MSE}(\hat{\theta}) > 0$:

$$P(|\hat{\theta} - \theta| > k\sqrt{\mathrm{MSE}(\hat{\theta})}) \le P((\hat{\theta} - \theta)^2 \ge k^2\mathrm{MSE}(\hat{\theta})) \le \frac{\mathrm{MSE}(\hat{\theta})}{k^2\mathrm{MSE}(\hat{\theta})} = \frac{1}{k^2}.$$

Taking complements proves (b).

(a) If $\mathrm{E}\hat{\theta} = \theta$, the bias–variance decomposition gives $\mathrm{MSE}(\hat{\theta}) = \mathrm{Var}(\hat{\theta}) = \sigma^2_{\hat{\theta}}$, so (a) is the special case of (b). With $k = 2$ the probability is at least $3/4$, the distribution-free statement made for confidence intervals above.

For a biased estimator, $\mathrm{MSE} = \sigma^2_{\hat{\theta}} + B^2 > \sigma^2_{\hat{\theta}}$, and the event $|\hat{\theta} - \theta| \le 2\sigma_{\hat{\theta}}$ need not have probability $3/4$: if $|B| > 2\sigma_{\hat{\theta}}$, then, e.g. for a normal $\hat{\theta}$, it has probability below $1/2$. The guarantee is restored only by measuring the error with $\sqrt{\mathrm{MSE}}$. ■

Exercise 1.2 (Birth weights: exact sampling distribution of $\bar{y}$)

For the four birth weights $5.5, 8.0, 6.5, 7.0$ of the example on sampling with and without replacement, let $\bar{y}$ be the mean of a sample of size $n = 2$.
(a) Under WOR, list the six sample means and compute $\mathrm{E}(\bar{y})$ and $\mathrm{Var}(\bar{y})$ by enumeration.
(b) Under WR (ordered pairs, each with probability $1/16$), compute $\mathrm{Var}(\bar{y})$ and the ratio of the two variances. Compare with σ^2/n and with $\frac{N-n}{N-1}\frac{\sigma^2}{n}$.

Solution. The population mean and variance are $\mu = 27/4 = 6.75$ and

$$\sigma^2 = \tfrac{1}{4}\left[(-1.25)^2 + 1.25^2 + (-0.25)^2 + 0.25^2\right] = \tfrac{3.25}{4} = 0.8125.$$

(a) The six samples, each with probability $1/6$, have means

$\{A,B\}$	$\{A,C\}$	$\{A,D\}$	$\{B,C\}$	$\{B,D\}$	$\{C,D\}$
6.75	6.00	6.25	7.25	7.50	6.75

Their sum is 40.5, so $\mathrm{E}(\bar{y}) = 40.5/6 = 6.75 = \mu$. The squared deviations from 6.75 are 0, 0.5625, 0.25, 0.25, 0.5625 so

$$\mathrm{Var}(\bar{y}) = \frac{1.625}{6} = 0.2708.$$

(b) Under WR the two draws are independent, each uniform on the four values, so $\mathrm{Var}(\bar{y}) = \frac{1}{4}\left[\mathrm{Var}(y_1) + \mathrm{Var}(y_2)\right] = \sigma^2/2 = 0.40625$. (Enumerating the 16 ordered pairs gives the same number.) The ratio is

$$\frac{0.40625}{0.2708} = 1.5 = \frac{N-1}{N-n} = \frac{3}{2}.$$

Thus the WR variance equals σ^2/n, and the WOR variance equals $\frac{N-n}{N-1}\frac{\sigma^2}{n} = \frac{2}{3}\cdot 0.40625$, in agreement with the general formulas of Ch. 2. Sampling without replacement is 50% more efficient here because half of the population is observed. ■

Exercise 1.3 (Shrinking the sample mean)

Let $y_1, \ldots, y_n$ be iid with mean $\mu \neq 0$ and variance $\sigma^2 > 0$, and consider $\hat{\theta}_c = c\,\bar{y}$ for a constant c.
(a) Find $\mathrm{MSE}(\hat{\theta}_c)$ as a function of c.
(b) Find the minimizing c^* and show that $\mathrm{MSE}(\hat{\theta}_{c^*}) < \mathrm{MSE}(\bar{y})$.
(c) Compute $\mathrm{RE}(\bar{y}, \hat{\theta}_{c^*})$ and discuss why $\hat{\theta}_{c^*}$ is not used in practice.

Solution. [Supplement]

(a) $\mathrm{E}(\hat{\theta}_c) = c\mu$ and $\mathrm{Var}(\hat{\theta}_c) = c^2\sigma^2/n$, so by the bias–variance decomposition

$$\mathrm{MSE}(c) = c^2\frac{\sigma^2}{n} + (c-1)^2\mu^2.$$

(b) $\mathrm{MSE}(c)$ is a quadratic in c with leading coefficient $\sigma^2/n + \mu^2 > 0$, and $\mathrm{MSE}'(c) = 2c\sigma^2/n + 2(c-1)\mu^2 = 0$ gives

$$c^* = \frac{\mu^2}{\mu^2 + \sigma^2/n} \in (0,1).$$

Substituting, with $v = \sigma^2/n$,

$$\mathrm{MSE}(c^*) = \frac{\mu^4 v + v^2\mu^2}{(\mu^2+v)^2} = \frac{\mu^2 v}{\mu^2 + v} < v = \mathrm{MSE}(\bar{y}),$$

the inequality being strict because $v > 0$.

(c) $\mathrm{RE}(\bar{y}, \hat{\theta}_{c^*}) = \mathrm{MSE}(\bar{y})/\mathrm{MSE}(c^*) = (\mu^2 + v)/\mu^2 = 1 + \sigma^2/(n\mu^2) > 1$, so the shrunken estimator is more efficient, and the gain is large when the coefficient of variation σ/μ is large relative to $\sqrt{n}$. However, c^* depends on the unknown ratio σ^2/μ^2; plugging in estimates gives a random c for which the bound of (b) is no longer guaranteed. The example shows that unbiasedness is a convention, not an optimality property: the MSE is the criterion that matters. ■